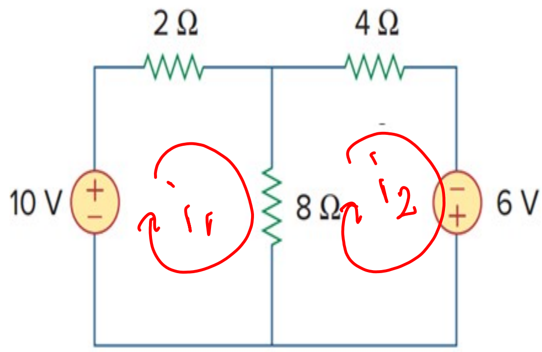
Problem 1

- Perform *branch current analysis* to determine the current absorbed by the $6\,V$ source in the following circuit.
- Perform *mesh analysis* to determine the current absorbed by the $6\,V$ source in the following circuit.



Applying KVL in mesh 1,

$$-10 + 2i_1 + 8(i_1 - i_2) = 0$$

$$\Rightarrow -10 + 2i_1 + 8i_1 - 8i_2 = 0$$

$$\Rightarrow \boxed{10i_1 - 8i_2 = 10} \text{ ——— (i)}$$

Applying KVL in mesh 2,

$$8(i_2 - i_1) + 4i_2 - 6 = 0$$

$$\Rightarrow 8i_2 - 8i_1 + 4i_2 - 6 = 0$$

$$\Rightarrow \boxed{8i_1 - 12i_2 = -6} \text{ ——— (ii)}$$

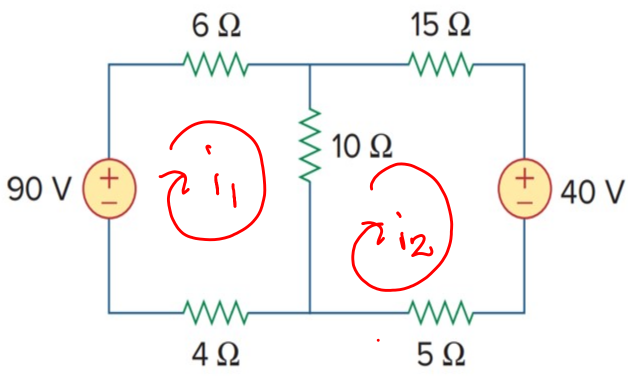
Solving (i) & (ii),

$$\boxed{i_1 = 3\,A} \qquad \boxed{i_2 = 2.5\,A}$$

Current absorbed by 6V source is (−2.5A)

Problem 2

- Calculate the current through the 10 Ω resistor using mesh analysis.



Applying KVL in mesh 1,

$$-90 + 6i_1 + 10(i_1 - i_2) + 4i_1 = 0$$

$$\Rightarrow -90 + 6i_1 + 10i_1 - 10i_2 + 4i_1 = 0$$

$$\Rightarrow \boxed{20i_1 - 10i_2 = 90} \text{ —— (I)}$$

Applying KVL in mesh 2,

$$10(i_2 - i_1) + 15i_2 + 40 + 5i_2 = 0$$

$$\Rightarrow 10i_2 - 10i_1 + 15i_2 + 40 + 5i_2 = 0$$

$$\Rightarrow \boxed{-10i_1 + 30i_2 = -40} \text{ —— (II)}$$

Solving (I) & (II),

$$\boxed{i_1 = 4.6A} \qquad \boxed{i_2 = 0.2A}$$

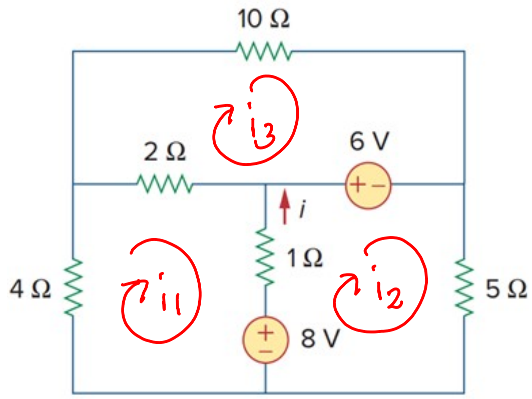
Current through 10 Ω Resistor

$$i_1 - i_2 = 4.6 - 0.2$$

$$= \boxed{4.4A}$$

Problem 3

- Calculate the current i using mesh analysis.



Applying KVL in mesh 1,

$$4i_1 + 2(i_1 - i_3) + 1(i_1 - i_2) + 8 = 0$$

$$\Rightarrow 4i_1 + 2i_1 - 2i_3 + i_1 - i_2 + 8 = 0$$

$$\Rightarrow \boxed{7i_1 - i_2 - 2i_3 = -8} \text{ ——— } (1)$$

Applying KVL in mesh 2,

$$-8 + 1(i_2 - i_1) + 6 + 5i_2 = 0$$

$$\Rightarrow -8 + i_2 - i_1 + 6 + 5i_2 = 0$$

$$\Rightarrow \boxed{-i_1 + 6i_2 = 2} \text{ ——— } (II)$$

Applying KVL in mesh 3,

$$10i_3 - 6V + 2(i_3 - i_1) = 0$$

$$\Rightarrow 10i_3 - 6V + 2i_3 - 2i_1 = 0$$

$$\Rightarrow \boxed{-2i_1 + 13i_3 = 6} \text{ ——— } (III)$$

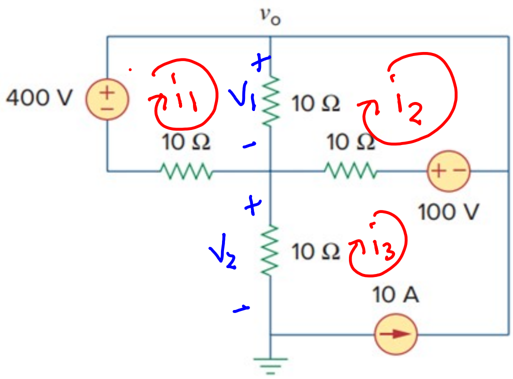
Solving (I), (II) & (III), $\boxed{i_1 = -1.03}$ $\boxed{i_2 = 0.16}$ $\boxed{i_2 = 0.30}$

$$i_o = (i_2 - i_1)$$

$$= 0.16 - (-1.03) = \boxed{1.19\,A}$$

Problem 4

- Apply mesh analysis to find v_o in the following circuit.



From the circuit,

$$\boxed{i_3 = -10\,A} \quad \text{(i)}$$

Applying KVL in the mesh 1,

$$-400 + 10(i_1 - i_2) + 10 i_1 = 0$$

$$\Rightarrow -400 + 10 i_1 - 10 i_2 + 10 i_1 = 0$$

$$\Rightarrow \boxed{20 i_1 - 10 i_2 = 400} \quad \text{(ii)}$$

Applying KVL in the mesh 2,

$$-100 + 10(i_2 - i_3) + 10(i_2 - i_1) = 0$$

$$\Rightarrow -100 + 10 i_2 - 10 i_3 + 10 i_2 - 10 i_1 = 0$$

applying $i_3 = -10\,A$,

$$-100 + 10 i_2 - 10 \times -10 + 10 i_2 - 10 i_1 = 0$$

$$\Rightarrow -100 + 20 i_2 - 10 i_1 + 100 = 0$$

$$\Rightarrow \boxed{-10 i_1 + 20 i_2 = 0} \quad \text{(iii)}$$

Solving (i) & (ii),

$$\boxed{i_1 = 26.67\,A} \qquad \boxed{i_2 = 13.33\,A}$$

From the circuit,

$$V_o = V_1 + V_2$$

$$V_1 = 10(i_1 - i_2)$$

$$= 10 \times (26.67 - 13.33)$$

$$= 133.4 \text{ V}$$

$$V_2 = 10 \times (-i_3)$$

$$= 10 \times (-(-10))$$

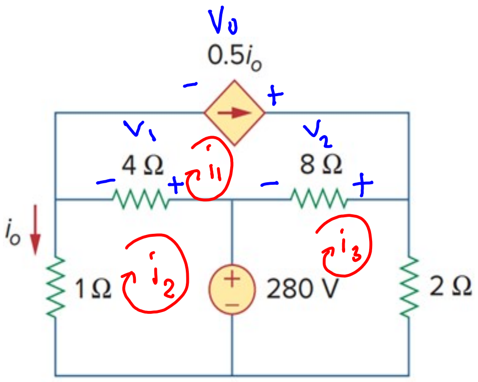
$$= 100 \text{ V}$$

$$\therefore V_o = 100 + 133.4$$

$$\boxed{V_o = 233.4}$$

Problem 5

- Find i_0 using mesh analysis. What is the voltage across the $0.5i_0$ source?



from circuit,

$$i_1 = 0.5 i_0$$

from mesh 2,

$$i_2 = -i_0 \Rightarrow i_0 = -i_2$$

$$\therefore i_1 = 0.5(-i_2)$$

$$\Rightarrow \boxed{i_1 + 0.5 i_2 = 0} \quad \text{—— (I)}$$

Applying KVL in mesh 2,

$$1 i_2 + 4(i_2 - i_1) + 280 = 0$$

$$\Rightarrow i_2 + 4i_2 - 4i_1 + 280 = 0$$

$$\Rightarrow \boxed{4i_1 - 5i_2 = 280} \quad \text{—— (II)}$$

Applying KVL in mesh 3,

$$-280 + 8(i_3 - i_1) + 2i_3 = 0$$

$$\Rightarrow -280 + 8i_3 - 8i_1 + 2i_3 = 0$$

$$\Rightarrow \boxed{8i_1 - 10i_3 = -280} \quad \text{—— (III)}$$

Solving (I), (II) & (III),

$$\boxed{i_1 = 20A}$$

$$\boxed{i_2 = -40A}$$

$$\boxed{i_3 = 44A}$$

let voltage across $2i_0$ source is V_0 &

$$V_0 = V_1 + V_2$$

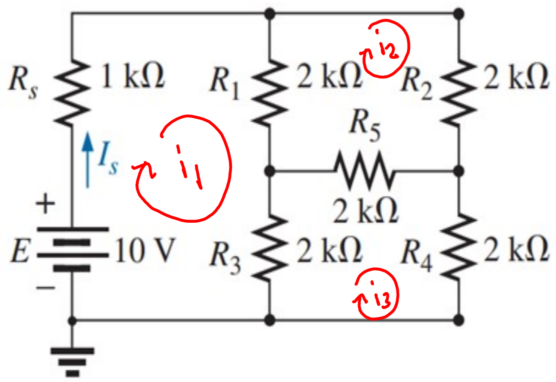
$$V_1 = (i_1 - i_2)4$$
$$= (20 - (-40))4$$
$$= 60 \times 4 = 240V$$

$$V_2 = (i_1 - i_3) \times 8$$
$$= (20 - 44) \times 8$$
$$= -192V$$

$$\therefore V_0 = V_1 - V_2 = 240 - 192$$
$$= \boxed{48V}$$

Problem 6

- Determine the current through the source resistor R_s using mesh analysis.



Applying KVL in mesh 1,

$$-10 + 1 \cdot i_1 + 2(i_1 - i_2) + 2(i_1 - i_3) = 0$$

$$\Rightarrow -10 + i_1 + 2i_1 - 2i_2 + 2i_1 - 2i_3 = 0$$

$$\Rightarrow \boxed{5i_1 - 2i_2 - 2i_3 = 10} \quad \text{(I)}$$

Applying KVL in mesh 2,

$$2(i_2 - i_1) + 2i_2 + 2(i_2 - i_3) = 0$$

$$\Rightarrow 2i_2 - 2i_1 + 2i_2 + 2i_2 - 2i_3 = 0$$

$$\Rightarrow \boxed{-2i_1 + 6i_2 - 2i_3 = 0} \quad \text{(II)}$$

Applying KVL in mesh 3,

$$2i_3 + 2(i_3 - i_1) + 2(i_3 - i_2) = 0$$

$$\Rightarrow 2i_3 + 2i_3 - 2i_1 + 2i_3 - 2i_2 = 0$$

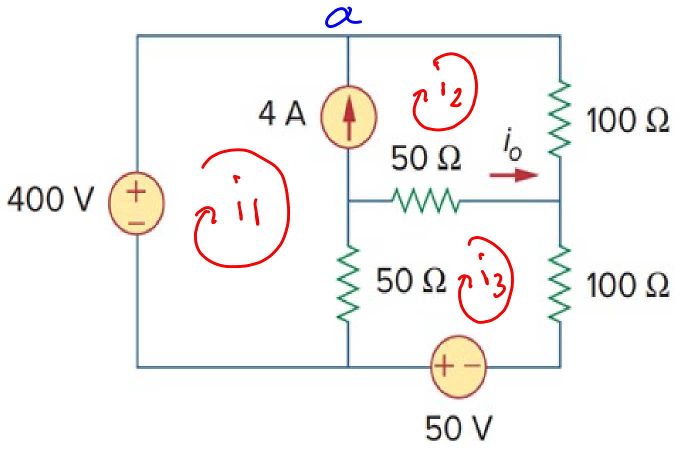
$$\Rightarrow \boxed{-2i_1 - 2i_2 + 6i_3 = 0} \quad \text{(III)}$$

Solving (I), (II) & (III),

$$\boxed{I_1 = I_S = 3.33\,A}$$

Problem 7

- Find i_0 using mesh analysis.



Applying KVL in mesh 3,

$$-50 + 50(i_3 - i_1) + 50(i_3 - i_2) + 100 i_3 = 0$$

$$\Rightarrow -50 + 50 i_3 - 50 i_1 + 50 i_3 - 50 i_2 + 100 i_3 = 0$$

$$\Rightarrow \boxed{-50 i_1 - 50 i_2 + 200 i_3 = 50} \text{ --- } (i)$$

Applying KVL in supermesh of 1 & 2,

$$-400 + 100 i_2 + 50(i_2 - i_3) + 50(i_1 - i_3) = 0$$

$$\Rightarrow -400 + 100 i_2 + 50 i_2 - 50 i_3 + 50 i_1 - 50 i_3 = 0$$

$$\Rightarrow \boxed{50 i_1 + 150 i_2 - 100 i_3 = 400} \text{ --- } (ii)$$

Applying KCL at node a,

$$i_1 + 4 = i_2$$

$$\Rightarrow \boxed{i_1 - i_2 = -4} \text{ --- } (iii)$$

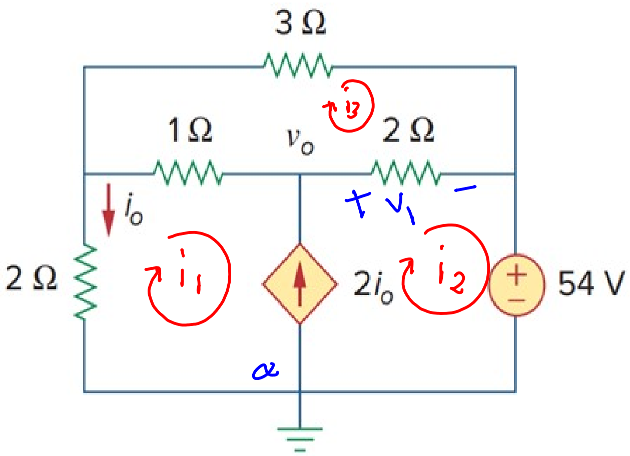
Solving (i), (ii), (iii),

$\boxed{i_1 = -0.5 A}$ $\boxed{i_2 = 3.5 A}$

$\boxed{i_3 = 1 A}$

$$i_0 = i_3 - i_2$$
$$= 1 - 3.5$$
$$= \boxed{-2.5 A}$$

Problem 8



- Find i_0 using mesh analysis. Determine the node voltage v_0.

Applying KVL in mesh 1,

$$3i_3 + 2(i_3 - i_2) + 1.(i_3 - i_1) = 0$$

$$\Rightarrow 3i_3 + 2i_3 - 2i_2 + i_3 - i_1 = 0$$

$$\Rightarrow \boxed{-i_1 - 2i_2 + 6i_3 = 0} \text{ ——— (I)}$$

Applying KVL in supermesh between mesh 1 & 2,

$$2i_1 + 1.(i_1 - i_3) + 2(i_2 - i_3) + 54 = 0$$

$$\Rightarrow 2i_1 + i_1 - i_3 + 2i_2 - 2i_3 + 54 = 0$$

$$\Rightarrow \boxed{3i_1 + 2i_2 - 3i_3 = -54} \text{ ——— (II)}$$

Applying KCL at node a,

$$2i_0 + i_1 = i_2$$

$$\Rightarrow 2(-i_1) + i_1 = i_2$$

$$\Rightarrow \boxed{i_1 + i_2 = 0} \text{ ——— (III)}$$

Solving (I), (II) & (III),

$\boxed{i_1 = -36A}$ $\boxed{i_3 = 6A}$

$\boxed{i_2 = 36A}$ $\boxed{i_0 = 36A}$

from the circuit,

$$V_0 = V_1 + 54$$

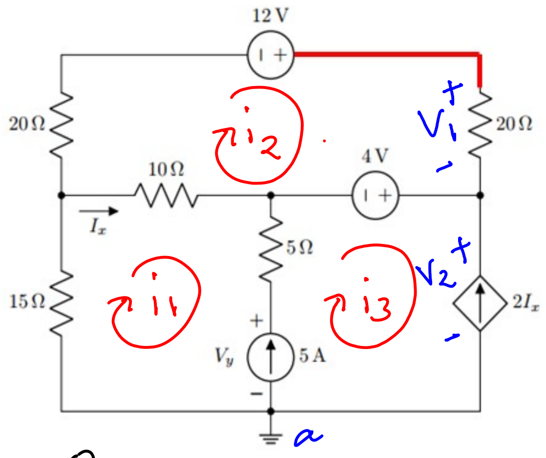
$$\Rightarrow V_0 = (i_2 - i_3)2 + 54$$

$$\Rightarrow V_0 = (36 - 6) \times 2 + 54$$

$$\Rightarrow \boxed{V_0 = 114V}$$

Problem 9

- Use Mesh Analysis to analyze the circuit. Find V_y. Determine the red colored node voltage.



Applying KVL in mesh2,

$$-12 + 20i_2 + 4 + 10(i_2 - i_1) + 20i_2 = 0$$

$$\Rightarrow -12 + 20i_2 + 4 + 10i_2 - 10i_1 + 20i_2 = 0$$

$$\Rightarrow \boxed{-10i_1 + 50i_2 = 8} \text{ ——— } (1)$$

from circuit,

$$i_3 = -2I_x$$

$$I_x = (i_1 - i_2)$$

$$\therefore i_3 = -2(i_1 - i_2)$$

$$\Rightarrow i_3 = -2i_1 + 2i_2$$

$$\Rightarrow \boxed{2i_1 - 2i_2 + i_3 = 0} \text{ ——— } (II)$$

Applying KCL in node a,

$$5 + i_1 = i_3$$

$$\Rightarrow \boxed{i_1 - i_3 = -5} \text{ ——— } (III)$$

Solving (I), (II) & (III),

$$\boxed{i_1 = -1.8A}$$

$$\boxed{i_2 = -0.2A}$$

$$\boxed{i_3 = 3.2A}$$

To find V_y, Applying KVL in mesh 1,

$$15i_1 + 10(i_1 - i_2) + 5(i_1 - i_3) + V_y = 0$$

$$\Rightarrow 15 \times -1.8 + 10(-1.8 + 0.2) + 5(-1.8 - 3.2) + V_y = 0$$

$$\Rightarrow \boxed{V_y = 68V}$$

To find red marked node voltage, $V_0 = V_1 + V_2$

$$V_1 = i_2 \times 20 = -0.2 \times 20 = -4V$$

To find, V_2, we apply KVL in the mesh 3,

$$-V_y + 5(i_3 - i_1) - 4 + V_2 = 0$$

$$\Rightarrow -68 + 5(3.2 + 1.8) - 4 + V_2 = 0$$

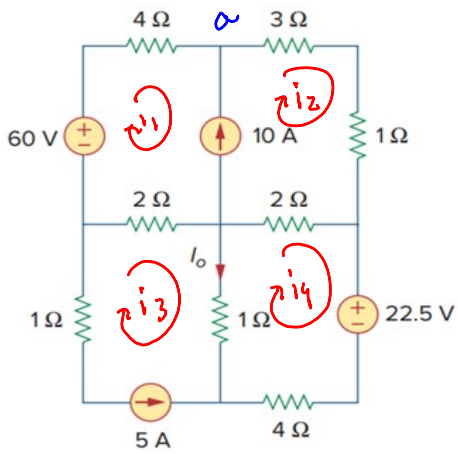
$$\Rightarrow V_2 = 47V$$

now, $V_0 = V_1 + V_2 = 47 - 4$

$$\Rightarrow \boxed{V_0 = 43V}$$

Problem 10

- Derive the mesh equations for the following circuit. Determine i_0.



Here, $i_3 = -5A$

Applying KVL in mesh 4,

$4i_4 + 1(i_4 - i_3) + 2(i_4 - i_2) + 22.5 = 0$

$\Rightarrow 4i_4 + i_4 + 5 + 2i_4 - 2i_2 + 22.5 = 0$

$\Rightarrow \boxed{-2i_2 + 7i_4 = -27.5}$ —— (i)

Applying KVL in supermesh between mesh 1 & 2,

$-60 + 4i_1 + 3i_2 + 1\cdot i_2 + 2(i_2 - i_4) + 2(i_1 - i_3) = 0$

$\Rightarrow -60 + 4i_1 + 3i_2 + i_2 + 2i_2 - 2i_4 + 2i_1 + 10 = 0$

$\Rightarrow \boxed{6i_1 + 6i_2 - 2i_4 = 50}$ —— (ii)

Applying KCL at node a,

$i_1 + 10 = i_2 \Rightarrow \boxed{i_1 - i_2 = -10}$ —— (iii)

Solving (i), (ii) & (iii),

$\boxed{i_1 = -1.06\,A}$ $\boxed{i_3 = -1.375\,A}$

$\boxed{i_2 = 8.94\,A}$

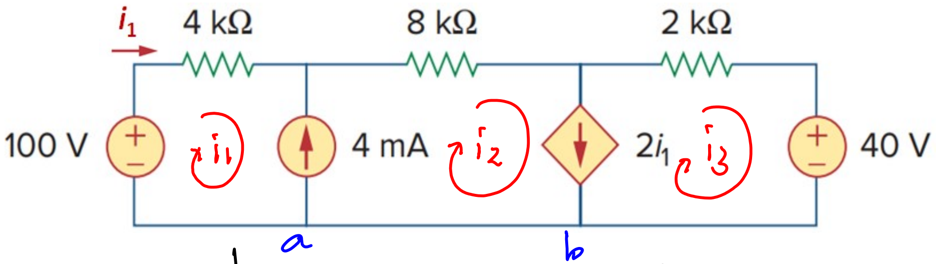
$i_0 = i_3 - i_4$

$\Rightarrow i_0 = -5 - (-1.375)$

$\Rightarrow \boxed{i_0 = -3.625}$

Problem 11

- Find the mesh currents.



Applying KVL in the supermesh among 1, 2 & 3,

$$-100 + 4i_1 + 8i_2 + 2i_3 + 40 = 0$$

$$\Rightarrow \boxed{4i_1 + 8i_2 + 2i_3 = 60} \longrightarrow (\mathrm{I})$$

Applying KCL at node a,

$$i_1 + 4 = i_2$$

$$\Rightarrow \boxed{i_1 - i_2 = -4} \longrightarrow (\mathrm{II})$$

Applying KCL at node b,

$$i_2 = 2i_1 + i_3$$

$$\Rightarrow \boxed{2i_1 - i_2 + i_3 = 0} \longrightarrow (\mathrm{III})$$

Solving (I), (II) & (III),

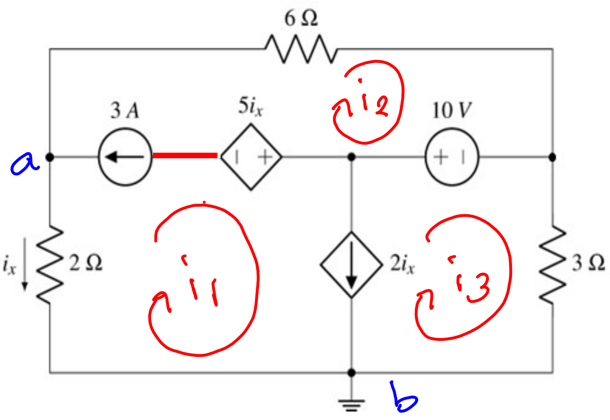
$$\boxed{i_1 = 2\,mA}$$

$$\boxed{i_2 = 6\,mA}$$

$$\boxed{i_3 = 2\,mA}$$

Problem 12

(i) Use nodal analysis to find i_x. Determine the voltage across the $3\ A$ source.
(ii) Use mesh analysis to find i_x. Determine the voltage of the red colored node



Applying KVL in supermesh among 1, 2 & 3,

$$\boxed{2i_1 + 6i_2 + 3i_3 = 0} \longrightarrow \text{(i)}$$

Applying KCL in node a,

$$i_1 + 3 = i_2$$

$$\Rightarrow \boxed{i_1 - i_2 = -3} \longrightarrow \text{(ii)}$$

Applying KCL in node b,

$$i_3 + 2i_x = i_1$$

$$\Rightarrow i_3 + 2(-i_1) = i_1$$

$$\Rightarrow \boxed{3i_1 - i_3 = 0} \longrightarrow \text{(iii)}$$

Solving (i), (ii) & (iii),

$$\boxed{i_1 = -1.059\ A}$$

$$\boxed{i_x = -i_1 = 1.059\ A}$$

$$\boxed{i_2 = 1.94\ A}$$

$$\boxed{i_3 = -3.176\ A}$$

Applying KVL in mesh 2,

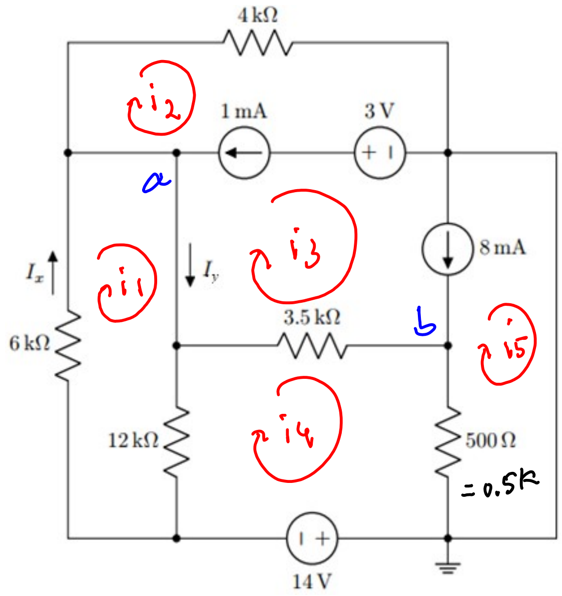
$$6i_2 - 10 + 5i_x - V_1 = 0$$

$$\Rightarrow 6 \times 1.94 - 10 + 5 \times 1.059 - V_1 = 0$$

$$\Rightarrow \boxed{V_1 = 6.935V}$$

Problem 13

- Use mesh analysis to analyze the circuit. Find I_x.
- Determine the current I_y.



Applying KVL in mesh 1,

$$6i_1 + 12(i_1 - i_4) = 0$$

$$\Rightarrow 6i_1 + 12i_1 - 12i_4 = 0$$

$$\Rightarrow \boxed{18i_1 - 12i_4 = 0} \text{ — (1)}$$

Applying KVL in mesh 4,

$$12(i_4 - i_1) + 3.5(i_4 - i_3) + 0.5(i_4 - i_5) + 14 = 0$$

$$\Rightarrow 12i_4 - 12i_1 + 3.5i_4 - 3.5i_3 + 0.5i_4 - 0.5i_5 + 14 = 0$$

$$\Rightarrow \boxed{-12i_1 - 3.5i_3 + 16i_4 - 0.5i_5 = -14} \text{ — (11)}$$

Applying KVL in supermesh of 2, 3, 5

$$4i_2 + 0.5(i_5 - i_4) + 3.5(i_3 - i_4) = 0$$

$$\Rightarrow 4i_2 + 0.5i_5 - 0.5i_4 + 3.5i_3 - 3.5i_4 = 0$$

$$\Rightarrow \boxed{4i_2 + 3.5i_3 - 4i_4 + 0.5i_5 = 0} \text{ — (111)}$$

Applying KCL at node a,

$$i_3 + 1 = i_2$$

$$\Rightarrow \boxed{i_2 - i_3 = 1} \quad \text{——— (iv)}$$

Applying KCL at node b,

$$i_5 + 8 = i_3$$

$$\Rightarrow \boxed{i_3 - i_5 = 8} \quad \text{——— (v)}$$

Solving (i), (ii), (iii), (iv) & (v),

$$\boxed{i_1 = -2mA}$$

$$\boxed{i_2 = -0.5mA}$$

$$\boxed{i_3 = -1.5mA}$$

$$\boxed{i_4 = -3mA}$$

$$\boxed{i_5 = -9.5mA}$$

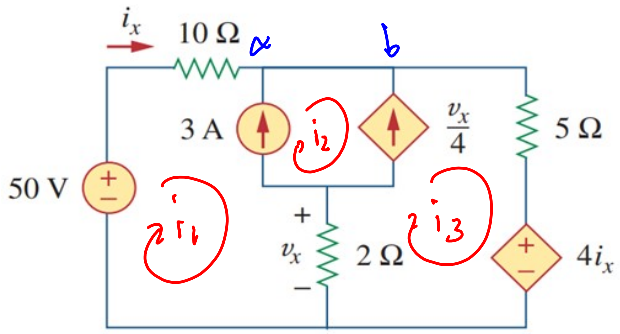
$$\boxed{I_x = i_1 = -2mA}$$

$$I_y = i_1 - i_3 = -2 - (-1.5)$$

$$\boxed{I_y = -0.5mA}$$

Problem 15

- Use mesh analysis to determine v_x and i_x. What is the voltage across the $3\ A$ source?



Applying KVL in supernode among 1,2,3,

$$-50 + 10i_1 + 5i_3 + 4i_x = 0$$

$$[i_x = i_1]$$

$$\Rightarrow -50 + 10i_1 + 5i_3 + 4i_1 = 0$$

$$\Rightarrow \boxed{14i_1 + 5i_3 = 50} \text{ —— (i)}$$

Applying KVL at point a,

$$i_1 + 3 = i_2$$

$$\Rightarrow \boxed{i_1 - i_2 = -3} \text{ —— (ii)}$$

Applying KVL at point b,

$$i_2 + \frac{V_x}{4} = i_3$$

where, $V_x = 2(i_1 - i_3)$

$$\therefore i_2 + \frac{2(i_1 - i_3)}{4} = i_3$$

$$\Rightarrow i_2 + 0.5i_1 - 0.5i_3 = i_3$$

$$\Rightarrow \boxed{0.5 i_1 + i_2 - 1.5 i_3 = 0} \longrightarrow (iii)$$

Solving (i), (ii) & (iii),

$$\boxed{i_1 = 2.105 A}$$

$$\boxed{i_2 = 5.105 A}$$

$$\boxed{i_3 = 4.105 A}$$

$$\boxed{i_x = i_1 = 2.105 A}$$

$$V_x = 2(i_1 - i_3)$$

$$= 2(2.105 - 4.105)$$

$$\boxed{V_x = -4V}$$